

IMMUNITY

Chapter

13

Major Concept

In this Unit you will learn:

- First Line of Defense (Skin, Digestive Tract, Air Passageway)
- Second Line of Defense - The Nonspecific Defenses
- Third Line of Defense - The Specific Defenses





Immunity:

No matter how much hygiene one keeps, our body is always exposed to pathogenic organisms such as viruses, bacteria, etc. They are present everywhere and have their own unique ways to infect our body. It's a matter of common observation that sometimes we feel abnormalities or disturbances in our normal systems, initially we use to bear and plan to visit some physician in the evening. Later, the complaint is resolved automatically and we forget what wrong early morning was. This miracle happens due to the immunity of our body which may be defined as "the resistance of our body against pathogenic living things and organisms". Due to the immunity, we not only check the entry of pathogens in our body but also the toxins they released after gaining access into our body. However, it's not the end of the defense tale, we do check and destroy our own abnormal cells as well as the foreign cells in our body.

Immunity is conferred to the animals through the activities of the **immune system**. It is a collection of cells and proteins that work in integrated fashion to protect the body from potentially harmful, infectious microorganisms. It is very important in the control of cancer, allergy, hypersensitivity and rejection problems when tissues and organs are transplanted. The study of the functioning and disorders of immune system is termed as **immunology**.

The immune system can be divided into following two functional divisions;

- (i) Innate Immune System, and (ii) Adaptive Immune System

(i) **Innate immune system:**

It's the part of our immune system which is provide us natural immunity. It is non-specific type as is against any pathogenic organism which tries to enter our body. It consists of two lines of defenses, viz., First and Second Lines of Defenses, respectively.

Table:13.1 Table summarizing our three lines of defenses

INNATE IMMUNE SYSTEM		ADAPTIVE IMMUNE SYSTEM
First Line of Defense	Second Line of Defense	Third Line of Defense
(NON-SPECIFIC)	(NON-SPECIFIC)	(SPECIFIC)
Physical Barriers: Intact Skin, Mucous Membrane, Ciliated Epithelium	Biochemical Barriers: Saliva, Mucus, HCl, Gastric Juice, Tears, Spermine(in semen)	Phagocytes, Natural Killer Cells, Inflammation, Antimicrobial Proteins, Fever.
		Cell Mediated Immunity; Antibody Mediated Immunity; B & T Cells

13.1 FIRST LINE OF DEFENSE:

Our first line of defense is considered to be non-specific in nature since it prevents the entry of all kinds of pathogens into our body. It is manifested by physical as well as biochemical barriers. The **physical barriers** as summarized in the (table 13.1) are skin, mucous membrane and ciliated epithelium. **The biochemical barriers** consist of saliva, tears, mucous, gastric juice, etc. Let's discuss their structure and role in protection one by one.

Skin- its structure and role in immune system

The skin is the one of the largest organ of our body. Being consists of 1.8 m^2 , it is a multitasking organ, playing its role not only in protection but also the regulation and sensation. It is always exposed to environmental hazards and stresses. The primary function of the skin is to act as a barrier. It provides protection from mechanical impacts and pressure, variations in temperature, micro-organisms, radiation and chemicals. It also regulates several aspects of physiology, including body temperature via sweat and hair; changes in peripheral circulation, and fluid balance via sweat. It does act as a reservoir for the synthesis of Vitamin D. As a sensory organ, it contains an extensive network of nerve cells that detect and relay changes in the environment. There are separate receptors for heat, cold, touch, and pain. The skin is composed of three main layers of closely packed cells. Among its layers are epidermis, dermis and hypodermis.

(i) Epidermis:

It is the outermost layer of skin. Since, it lacks blood supply, it is entirely dependent upon dermis for the supply of nutrients and disposal of wastes. It is made up of closely packed dead cells with keratin which makes the skin's surface mechanically tough and resistant to degradation by bacterial enzymes. Fatty acids on the skin's surface create a dry, salty, and acidic environment that inhibits the growth of some microbes and is highly resistant to breakdown by bacterial enzymes. In addition, the dead cells of the epidermis are frequently shed,

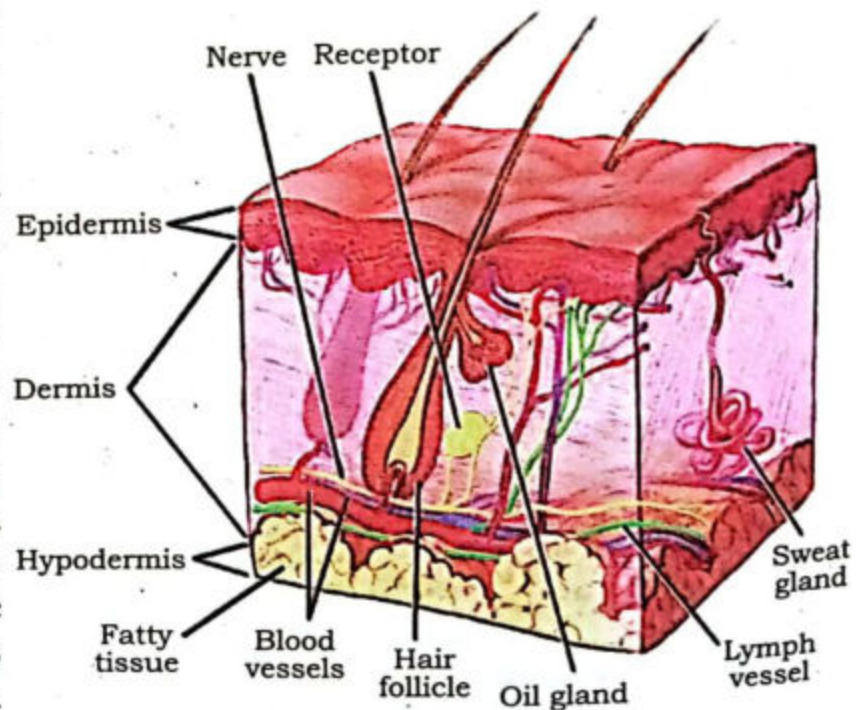


Fig: 13.1
3D image of human skin



along with any microbes that may be clinging to them. Shed skin cells are continually replaced with new cells from below, providing a new barrier that will soon be shed in the same way. Wandering among the three layers of skin, there is a variety immune cells such as **Langerhans Cells** (skin resident macrophages) and T cells for the elimination of intruding microbes.

(ii) **Dermis:**

Underneath the epidermis, the layer of skin is called dermis. It consists of living cells, blood vessels and nerves. It has sweat secreting **sweat glands** and sebum (oil like substance) secreting **sebaceous glands**. The later opens into hair follicles to lubricate the hair. The sebum contains some acids which make a thin film on the skin surface. It acts as chemical barrier for microorganisms to penetrate through it. On the other hand, sweat glands do contribute to this chemical barrier thereby secreting variety of polypeptides for the inhibition of the growth of microorganisms on the skin surface.

(iii) **Hypodermis:**

Below the dermis lies the hypodermis. It contains mainly fat storing adipose tissue and connective tissue. It connects the skin with the underlying muscles and bones. It serves as insulating layer.

Digestive Tract: Role in immune system

Our gut is lined internally with mucosa. In addition, our stomach has gastric glands which secrete HCl and variety of enzymes to destroy intruding noxious microorganisms. The internal lining of mouth, nose, digestive tract, respiratory passage and lungs, urinary tract, etc., contains mucous secreting cells hence is also known as **mucous membranes** or **mucosa**. These internal linings are made up of epithelial cells. The mucosa not only serves as non-specific physical barrier due to its compact nature but also provides a biochemical barrier through its mucous which contains many antimicrobial peptides. Moreover, the mechanical action of the peristalsis of the gut also moves out the sloughed mucous along with the undigested food (feces).

In addition to the above mentioned defenses, our digestive tract is also equipped with substantial amount of lymphoid tissue being located into three sectors, viz. **Tonsils** in pharyngeal region; **Peyer's patches** in small intestines and the **Appendix**. The lymphoid tissue is rich in macrophages and lymphocytes for the protection against pathogenic microorganisms which they encounter. Another sector of the lymphoid tissue are the W.B.Cs (e.g., Plasma Cells and Lymphocytes) wandering in the basement membrane of the small intestine. A number of lymphocytes also present in the epithelial

cells of mucosa offer another protection against the pathogenic microorganisms.



Fig: 13.2 Microscopic cross section of Human Small Intestine showing Peyer's Patches

Air Passage Way: Role in first line of defense:

In some systems like respiratory system, where there is a great risk of entry of pathogenic microorganisms through air, the mucosa traps the microorganisms and its ciliated surface flushes off the contaminated mucous outside so as to avoid retention of dust, dirt, pathogenic microorganisms inside thereby playing important role in first line of defense.

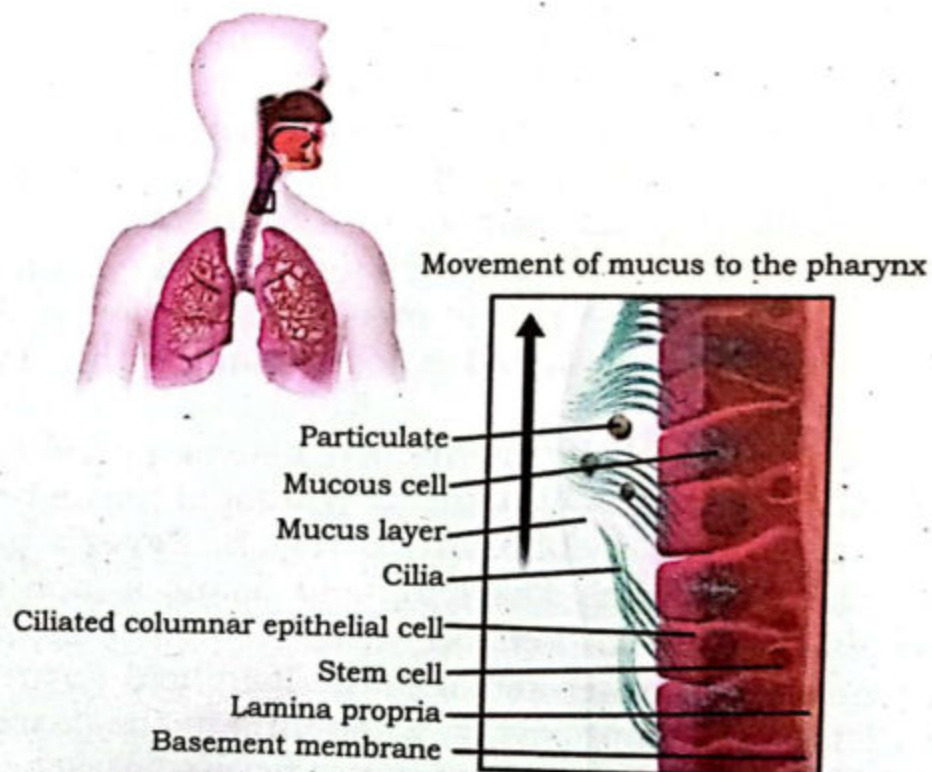


Fig: 13.3 Human respiratory epithelium



The hairs present in our nose do trap the dust, dirt, etc., from the inhaled air thus make it sure that the air passing into the rest of the air passage ways should be free of such contaminations.

The mucous membrane of the respiratory system is composed of ciliated columnar epithelial cells layer. It contains numerous mucous secreting mucous glands. Due to mucous, the dust and germs stick to it which is removed by the synchronized upward movement of cilia towards the pharynx. It is either pushed into the esophagus or removed through the mouth.

13.2 SECOND LINE OF DEFENSE (The Non-specific Defense)

The pathogenic microorganisms may enter our body when it is broken either accidentally like cut, burnt, etc. or through insect bite. Thus, if somehow the first line of our defense is breached, we do have the second line of defense to encounter. Like first line of defense, it is also non-specific in nature and comprised of a wide variety of our WBCs and defensive proteins. It is accomplished by the following major responses: Phagocytosis, cytotoxicity, inflammation and fever.

13.2.1 Killing Cells of Blood:

Whenever some pathogen gets access into the body through wound, broken skin, inhaled air, etc., it encounters WBCs, some of which recognize and kill it either by phagocytosis or direct killing. Let's learn them one by one.

(i) **Phagocytosis:**

It is the process through which some specific types of WBCs take-in and digest the foreign particle or microorganism. Among the WBCs, neutrophils, macrophages (monocytes) and dendritic cells perform such phagocytosis.

Neutrophils:

These are the most abundant WBCs (50%-70% of all leukocytes) and are specially adapted to phagocytize bacteria. The mature neutrophils have multi-lobed (3-5 lobed), non-spherical nucleus. They are usually short-lived. Upon maturation, majority of them migrate from the blood into our tissues. Through specific receptors on their surface, the neutrophils (figure, 13.5) can recognize various bacterial molecules such as peptidoglycans, flagellin protein of flagella, lipopolysaccharides, lipopeptides, etc.

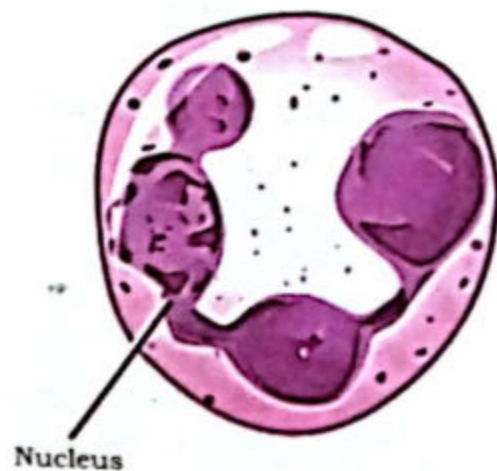


Fig: 13.4 A neutrophil

Upon recognition as foreign, the intruder is internalized by neutrophils and digested to destroy through its lysosomal hydrolytic enzymes.

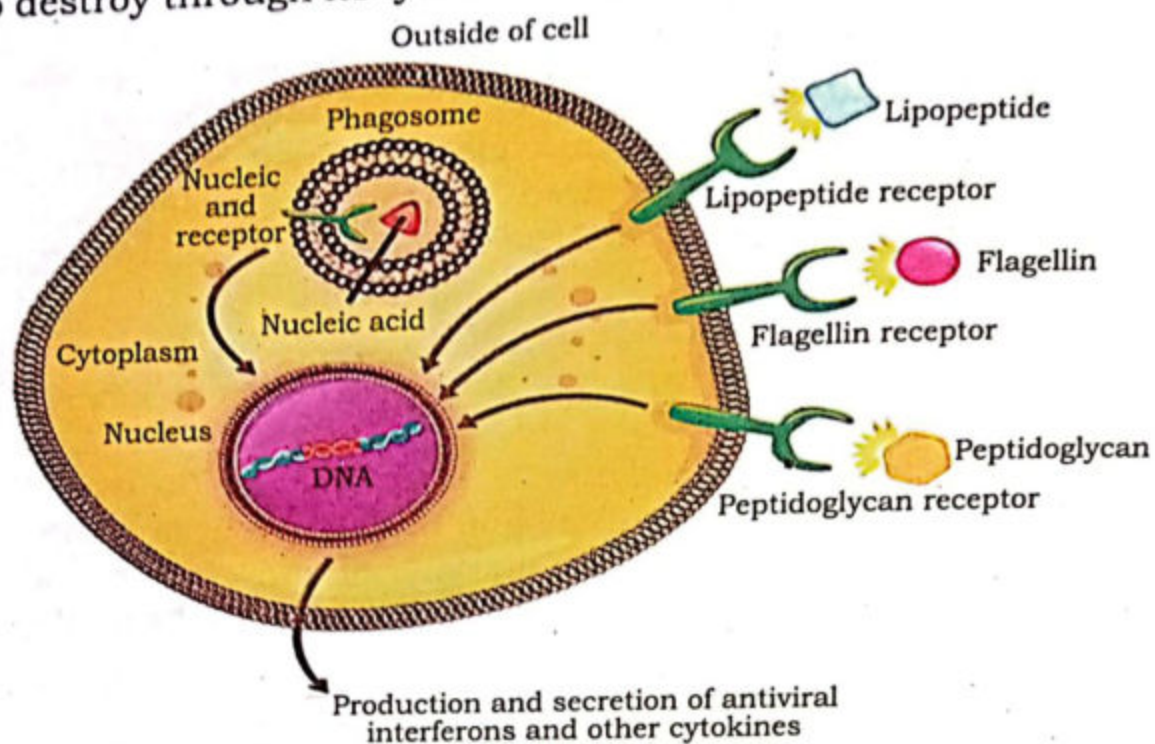


Fig: 13.5 Phagocytosis of bacteria

Monocytes:

They have a single bean shaped nucleus. The cytoplasm is non-granulated. They constitute about 2-8% of our total W.B.Cs. Like granulocytes, they originate from the myeloid stem cells. Although, the average life span is about 2 to 5 days in blood circulation, they can migrate from the blood into the tissue where they grow in size and become either **macrophages dendritic cells** and can live for months to years. Monocytes are involved in our innate immunity.

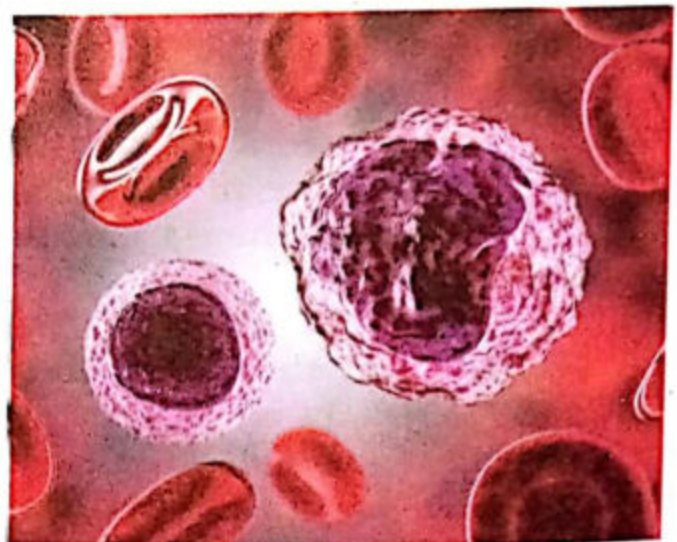


Fig: 13.6. A monocyte

Macrophage:

The leukocytes other than neutrophils involved in phagocytosis of foreign cells, worn-out cells, etc. are macrophages. They are derived from monocytes which are categorized as a type of agranulocytes because they lack any kind of pigments in their cytoplasm. They are the largest among all leukocytes. Unlike granulocytes, they have a large indented, horse-shoe shaped, nucleus. Circulating monocytes, after sometimes leave blood and



enter into lymphoid as well as non-lymphoid tissues where they grow in size and turn into macrophage dendritic cells. Macrophages have ability to adapt the specific tissue conditions so they are also named such as Alveolar macrophages in alveoli of lungs, Kupffer cells in liver, Microglia in Central Nervous System, etc. accordingly.

Macrophages not only phagocytize the pathogenic microorganisms but also remove and present their antigens to T-Cells (a type of lymphocytes), hence they are also termed as **Antigen Presenting Cells (APCs)**. They do release specific signaling molecules called **cytokines** to activate other immune cells. They also initiate the process of inflammation which we will discuss in the proceeding text. In addition to above mentioned activities, macrophage secrete nitric oxide which kills phagocytized pathogenic organisms.

Dendritic Cells:

The dendritic cells, another type of WBC, are also involved in presenting antigens of the phagocytized pathogen to the T-Cells. They were named so due to the presence of long cytoplasmic projections or dendrites. Upon activation through specific inflammatory cytokines secreted by macrophages, the dendritic cells migrate to the secondary lymphoid tissues such as tonsils, Peyer's Patch, spleen, tonsils, etc.). As soon as their task of activation of T cells achieved, unlike macrophage they die.

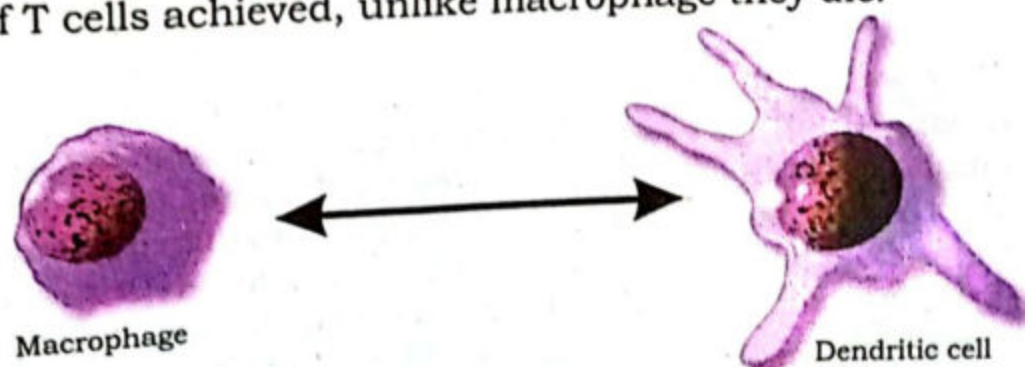


Fig: 13.7 Macrophage vs Dendritic Cell

(ii) Cytotoxicity:

It refers to the ability of a substance or certain cell, either to damage or cause death of a target cell. It is one of our protective response is exhibited against virally infected cells, cancer to tumor cells as well as our cells which are being infected by some pathogenic microorganism. The process of cytotoxicity could be a consequence of either Natural Killer (NK) Cells or Cytotoxic T (T_c) Cell. Although both NK Cells as well as T_c Cells are lymphoid in cell lineage and originate in the bone marrow, yet they differ in their mode of activation against their target cells.

Natural Killer Cells (NK Cells):

NK Cells are one of the types of lymphocytes involved in detecting and eliminating virally infected cells as well as cancer cells. In such cases, unlike T_C cells, they do not need any prior activation by antigen presenting cells.

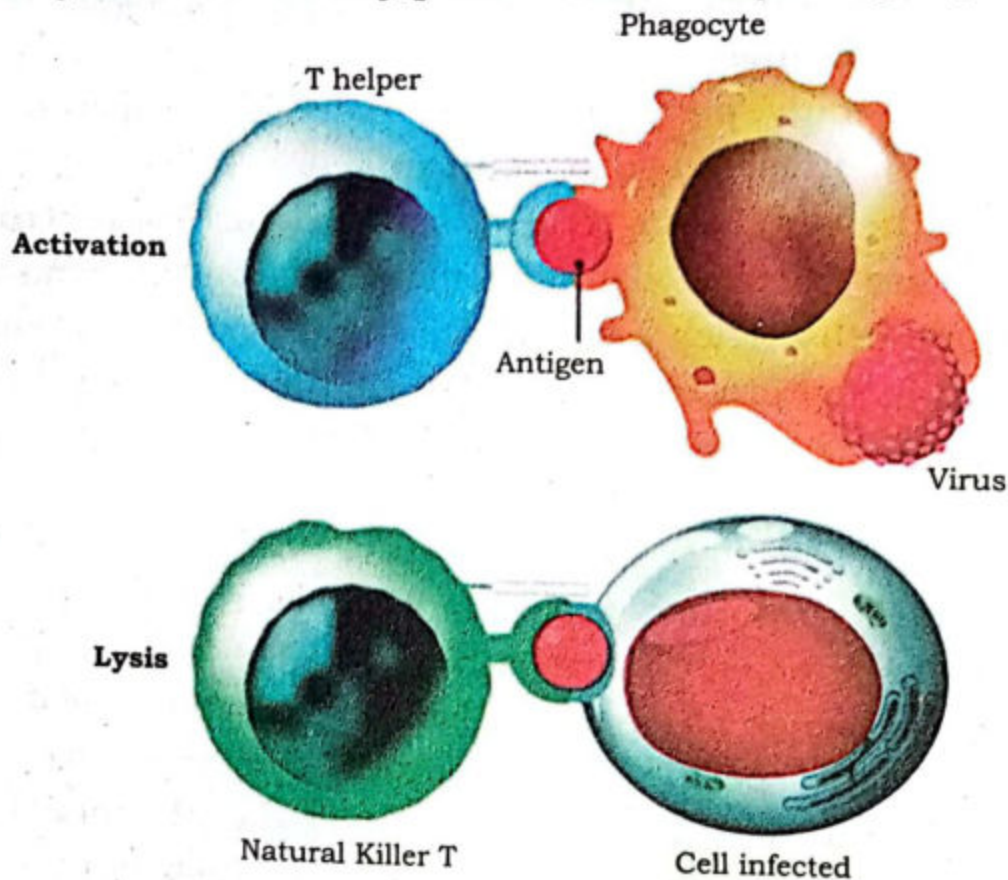


Fig: 13.8. Response of NK Cell against a normal vs infected cell. In case of infected cell it is activated to cause lysis of the target cell.

Hence they were named as Natural Killer Cells. However, the mode of action in destroying the target cells is almost similar. Just like a policeman checking national Identity card of person for verification of his identity, the NK Cells do check on the cell surface, a conjugate molecule called Major Histocompatibility Complex-I (MHC-I), which serves as identification card of cell as "self". In case of intact MHC-I, the NK cell remain inactive while in case of any modification or absence of MHC-I on the target cell, it is activated to destroy the target cell. The lysis of the target cell is brought about by releasing cytotoxic granules having **perforin** protein and **granzyme**. The former forms pores in the plasma membrane of the target cells while the later enters the target cell and causes its lysis.

Cytotoxic T Cells (T_C Cells):

Cytotoxic T Cells are another group of immune cells which belong to Lymphocyte family of our W.B.Cs. As the name suggests, they recognize virally infected cells as well as tumor cells. Though they are produced in the bone marrow but they migrate to thymus where they mature and then released into the blood. During their maturation process, the nive T Cells



which develop CD8 type of co-receptor along with their T Cell Receptor (TCR) on their surface, turn into Cytotoxic T Cells. Through their CD8 co-receptor and TCR, the Cytotoxic T Cells bind with **Major Histocompatibility Complex-I (MHC-I)** present on all of their own nucleated cells of our body except macrophages because they have another **Major Histocompatibility Complex-II (MHC-II)**. For the activation of Tc cells, several chemical signals are required from other cells like dendritic and T helper cells.

Whenever the Tc cells encounter virally infected cells, they recognize them immediately through the viral antigens expressed on the MHC-I of the infected cells. Their programming, in such case, is to release some protective proteins or cytokines such as Tumor Necrosis Factor – alpha (TNF- α) and Interferon-gamma (IFN- γ) which act on other immune cells such as macrophage and dendritic cells to enhance immune responses. Like NK Cells, they release **perforin** proteins and **granzyme** to cause lysis of the infected target cell. Unlike NK Cells, the Tc cells then proceed towards other infected cells to cause their lysis. In this way, they serve the role as “serial killer”.

13.2.2. Protective proteins:

In our body fluids, there circulates different proteins which help us in the protection of the body against microorganisms such as pathogenic bacteria, etc. Such proteins are termed as complement protein. Upon recognition of bacteria as foreign organism (non-self), complement proteins are activated through a series of reactions. Some of them damage the plasma membranes of their target cells while the others attract other cells of the immune system such as mast cells, neutrophils and macrophages to enhance the immune responses.

13.2.3. Inflammatory response:

Inflammation (to set on fire) is one of the non-specific immune responses which is usually a consequence of infection. However, it may be due to other factors like tissue damage, irritants or autoimmune disorders. It is classically categorized as two types, acute and chronic. The acute type is a short term inflammation and characterized by redness, hotness, pain, swelling and loss of the function of the affected organ. The chronic one is of long term duration and is usually a consequence of diseases like Cardio-Vascular Disorder (CVD), allergy, etc.

Whenever, damage to our tissue takes place either through an accident or a pin prick to skin, the harmful bacteria are introduced in our tissues. As a result, the damaged tissue releases specific complex proteins such as Histamine and Prostaglandin which dilate the nearby blood vessels and attract neutrophils and macrophages to the site of infection so that the intruder bacteria may be eliminated through phagocytosis. It may result in abscess. Since it is a sort of war is going on in the infected tissue, our own

cells may die forming a thick paste-like, white, yellow, orange or greenish fluid, Pus is formed. The pus may have bad smell.

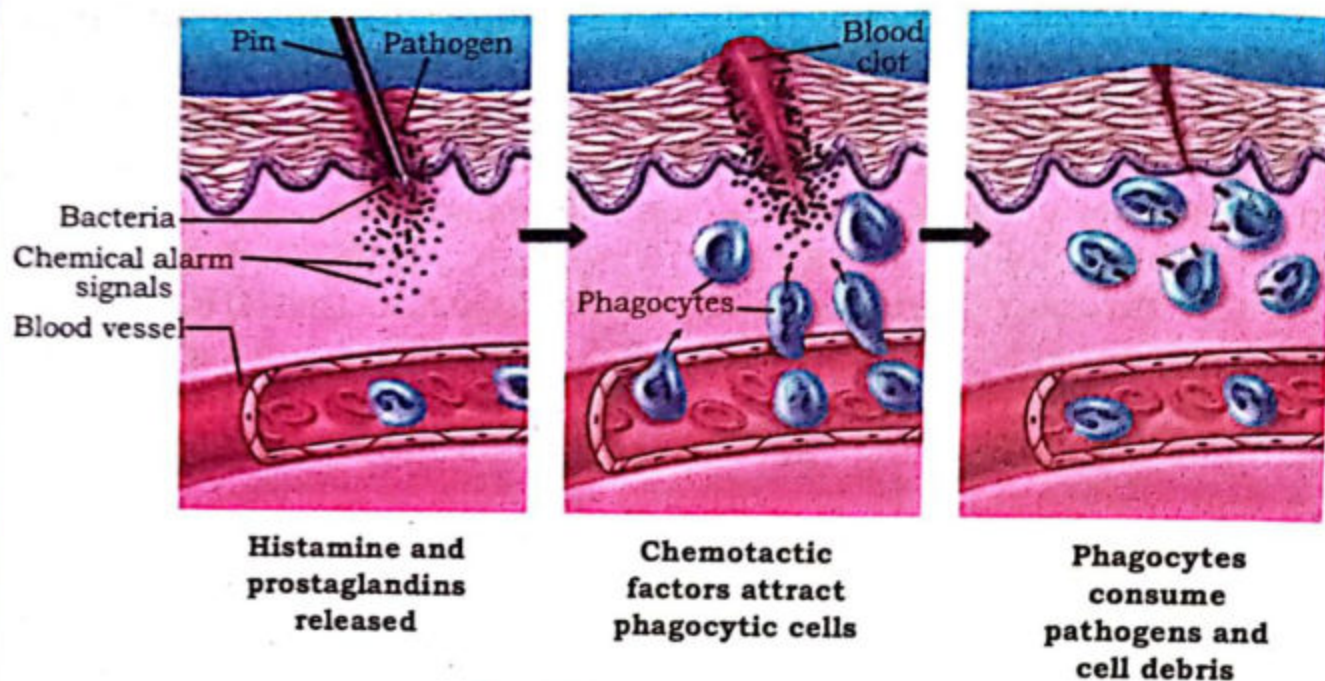


Fig:13.9. Inflammatory response

Justify the inflammatory response in arthritis as an example of a misdirected immune response:

Normally the immune cells respond to the foreign cells, or particles such as bacteria, viruses, etc. only. However, in few cases like autoimmune disorder such as Rheumatoid Arthritis, the immune cells respond against their self (own) cells/antigens. The exact reasons for this misdirection are not certain in many specific cases. In case of rheumatoid arthritis, there is inflammation of the synovial membranes of the joints that progresses to swelled and painful joint, ultimately leading to their deformity. It may damage other parts of the body also.

13.2.4 Temperature response or fever (Pyrexia):

Whenever the body temperature rises above the normal, it is clinically termed as fever. It is a symptom rather than a disease. It is one of the non-specific type of our immune system which could be a sign of infection. It may also be caused by heat-stroke, brain tumors, toxins affecting brain, some medicines, etc. Substances that induce fever are termed as **Pyrogens**. They may be endogenous (body's own cells secretions) or exogenous (external source). The endogenous pyrogens are released by our immune cells in response to the presence of pathogenic organisms like bacteria, viruses, etc. They are released in the blood stream from actively phagocytizing macrophages. The important pyrogens are Interleukin-1 (IL-1) and Interleukin-6 (IL-6), Tumor Necrosis Factor (TNF) and Interferons (INF-alpha), respectively. Through the blood, the pyrogens are transported to the brain where they act upon hypothalamus, the thermostat of the brain. As a



consequence, the hypothalamus produces lipid like signaling molecules, **prostaglandin** due to which the thermogenesis is increased and heat loss is decreased. This causes rise in the body temperature or fever.

Because of the fever, many fungi and bacteria cannot grow and multiply properly so that our immune cells can handle them easily causing their death, e.g. facilitates the phagocytosis. As mentioned earlier, the exogenous pyrogens are coming from external source such as bacteria. Such toxins are recognized by brain as pyrogens and can induce fever.

Although fever in the above sense is beneficial to us, yet it has following harmful effects also:

- i) A considerable amount of energy is lost as heat.
- ii) It causes fatigue, dehydration, body ache and seizures also.
- iii) Temperature higher than 105°F denature our enzymes and other proteins also.
- iv) It can denature our own cells. It can cause death also.

Justify why physicians prescribe antipyretic drugs, when fever is non-specific defense against microbial infections.

Antipyretic Drugs: Those drugs which reduce or control fever are called antipyretic. The antipyretics can inhibit the prostaglandin secretion by the hypothalamus thereby reducing fever. The antipyretics can be categorized into 3 groups, viz., i) Salicylates, e.g. Aspirin, etc. ii) Acetaminophen, e.g. Paracetamol, etc. and iii) Non-steroidal anti-inflammatory, e.g. Ibuprofen, etc. Most of the antipyretics inhibit enzyme, cyclooxygenase which reduces the level of prostaglandins from the brain.

13.3 THIRD LINE OF DEFENSE (The Specific Defenses):

The non-specific second line of our defense is supported simultaneously by third line of defense. It is very complex and specific in nature. This part of the immune system is termed as Acquired or Adaptive Immune System and its response varies from person to person and is the most powerful means of resisting infection.

Recognition of self vs non-self:

For the adaptive immune response to work, it is first necessary to recognize the foreign organism as "non-self" by the immune cells. It is made possible by the identification of **Antigens** of invading organism. The antigen could be any foreign macromolecule, especially proteins which can elicit immune response. In response to the particular antigens, B-lymphocytes produce a small, soluble and specific protein called **antibody** which combines with specific antigen and helps to destroy the antigen displaying pathogenic organisms. The immune system of a vertebrate can produce billions of different antibodies. Not only the immune system destroys in this

way the pathogenic organism but it also memorizes it and can produce antibodies against the same pathogen in case of reinfection. The distinction between the self and non-self is the hallmark of immune system.

As you have already learnt about the Major Histocompatibility Complex (MHC) Class-I & Class-II as our cell surface marker or Cell Identification Cards which are to be displayed on cell membranes of each and every cell of our body. Being codes by a specific set of genes, MHC proteins are found in all vertebrates. In humans, they are also termed as Human Leukocyte Antigens (HLA) system. As you can see in (Figure 13.10) that the MHC-I is composed of three alpha sub-units and a beta microglobulin. In between alpha 2 and alpha1, there is groove termed as binding groove for interaction with NK Cells through its inhibitory receptor. On the other hand, the MHC-II is comprised of 2-unit alpha chains and 2-unit beta chains (Heterodimer). The binding groove is shallow and open at each ends.

MHC-I is present on the plasma membrane of our all nucleated cells and platelets. It interacts with T_c cells and NK Cells. In case of intact MHC-I, the patrolling cells take it as "self-cells" and no adverse response is shown against them. In case of any kind of alteration/modification/loss in MHC-I, the target cell is taken as non-self and subject to be terminated. Thus altered MHC-I serves as a global alarming system in our body's immune system.

MHC-II is displayed on the cell surface of antigen presenting cell such as macrophage, dendritic cells and B cells. The MHC-II interacts with CD4 T cell, i.e., Helper T Cell and is essential for our adaptive immune system so as to develop

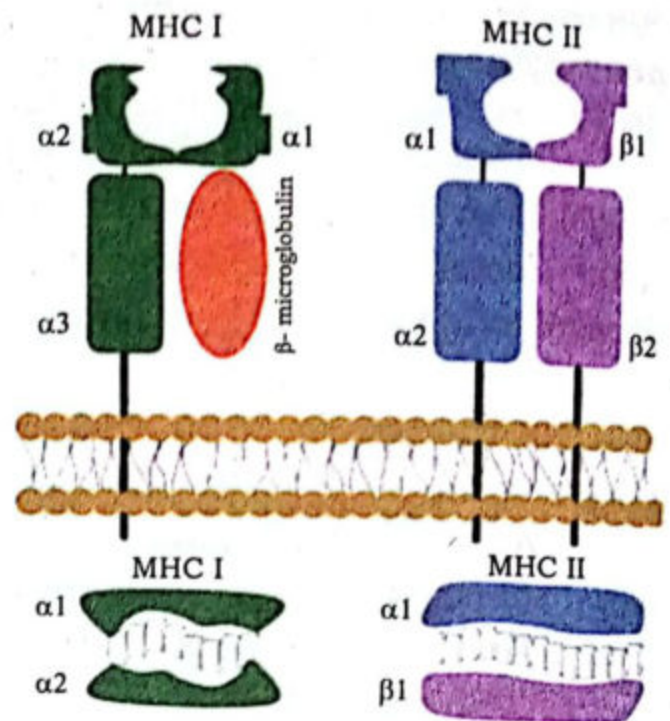


Fig: 13.10. MHC peptides expressed on the cells surface.

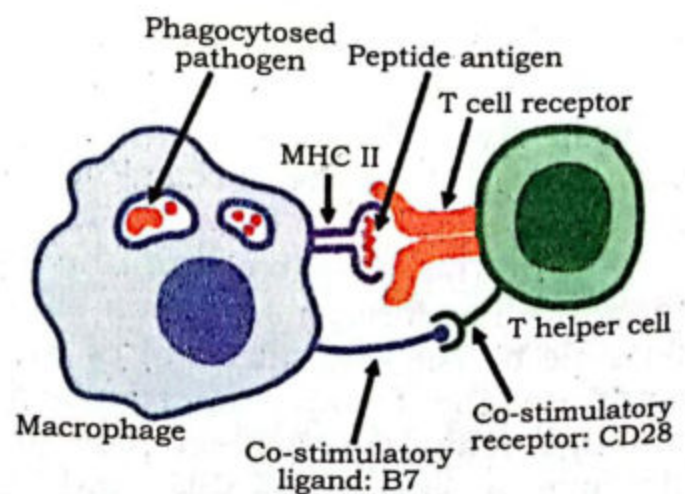


Fig: 13.11 Interaction of an antigen presenting cell with T helper cell



specific immune response. If no peptide antigen is displayed on MHC-II of macrophage, the T_H cell remain inactive. However, in case of such antigen displayed on MHC-II of APC, the naïve T_H cell activates and releases number of lymphokines to activate other immune cells.

13.3.1 In-born and acquired immunity:

There are two basic types of immunity, Innate Immunity and Acquired Immunity.

Inborn or Innate Immunity:

The Inborn or innate immunity provides the non-specific, natural defense of our body against pathogenic organisms. It is conferred upon us through physical barriers (e.g. skin, etc.), cellular barriers (e.g. macrophage, neutrophils, mast cells, eosinophils, and basophils), biochemical barriers (e.g. cytokines, interferon, interleukins etc.). This kind of immunity is a consequence of genetic make-up of an individual so it exists before the infection is acquired.

Acquired or Adaptive Immunity:

After the breach of the Inborn immunity, the acquired or adaptive immunity comes in action against pathogenic or foreign organisms. It is specific in nature and the pathogen is death with T and B lymphocytes. Although the resistance offered is slower, it is long lasting due to existence of memory, and in few cases life long against particular pathogens. In this case the resistance offered by individual varies from person to person.

There are two types of acquired immunity: i) Active Immunity, and ii) Passive Immunity. Both of these could be natural or artificial. If immunity develops as a consequence of natural infection, it is termed as Natural, and if it is a consequence of artificially inoculation of pathogenic organism, it is termed as Artificial, as shown in the table number 13.2). Artificial immunity is also termed as Immunization.

Table: 13.2 Innate and Artificial Immunity

	Active Immunity	Passive Immunity
Natural	Clinical, sub-clinical infection	via breast milk, through placenta
Artificial	Vaccination: Live attenuated, killed pathogens; purified antigen vaccine	Inoculation of immune serum, immune cells from other individual or other organisms.

(i) Active Immunity:

It is a consequence of individuals' own immune response. If it is due to natural infection, it is called Natural Active Immunity. In case of artificially inoculating individual with vaccine, it is termed as Artificial Active Immunity or Vaccination.

Vaccine:

A vaccine is either a pathogen (live attenuated, or killed) or its product that is introduced in our body to induce a state of immunity for protection against natural infection with the same pathogen. Vaccines are of following four types:

- i) Live Attenuated Vaccines,
- ii) Inactivated Vaccines,
- iii) Toxoid Vaccines, and
- iv) Sub-unit, Recombinants, polysaccharide and Conjugate Vaccines.

(ii) Passive Immunity:

In this kind of immunity, antibodies or immune cells produced by one individual (donor) are transferred to another individual (receptient) to develop immunity. For example, a pregnant woman passes her antibodies to her foetus through placenta. Similarly, an infant receives antibodies of his/her mother through breast feeding. This kind of passive immunity is termed as Natural Passive Immunity. There could be another case of passive immunity in which immune cells or immune serum from an immune person are transferred to non-infected person. This is termed as Artificial Passive Immunity.

13.3.2 Cell mediated immunity and antibody mediated immune responses:

The third line of defense is based upon our specific W.B.Cs. known as Lymphocytes. They are of two types, i.e., T cells and B cells, respectively. Although both are morphologically indistinguishable from each other, yet they differ a lot biochemically and functionally.

B cells mature in bone marrow and then are released into the blood circulation while the T cells are released in immature-form from the bone marrow, from where they migrate to thymus and become mature. After maturation they are released into the blood circulation. Since they mature in thymus, they are termed as T cells. B cells were named as they were discovered in a glandular sac opening called Bursa of Fabricius into the cloaca of chicken. The third line of defense or Adaptive Immune System provides us immunity in two different ways, i.e., Cell Mediated Immunity (CMI) through T cells and Antibody Mediated/Humoral Immunity through B Lymphocytes, respectively.

(i) Cell mediated immune (CMI) response:

It is a consequence of T-Lymphocytes which do not secrete antibodies. The CMI is particularly important against viruses, parasites that hide within the host cells, tumors cells and fungi.



(ii) **Antibody mediated immune response (Humoral Immunity):**

It takes place due to the activity of another group of lymphocytes, the B-Cells. It was once called "humoral" because the antibodies are secreted in the blood stream or humours.

In order to learn about CMI & Antibody mediated Immunity, we will some structural, biochemical and functional details about lymphocytes.

Lymphocytes:

The lymphocytes are very important group of W.B.Cs which originate from lymphoid stem cell series. They constitute 20-40% of the circulating W.B.Cs and play crucial role in adaptive immune system. They are found in blood as well as tissues. There are two types of lymphocytes, T Cells and B Cells. Both are antigen-specific and have special receptors called T Cell Receptors (TCR) and B Cell Receptors (BCR) on their plasma membranes, respectively. Their average life span is about a week to few months but a few may live for years. The long-lived lymphocytes are actually those which confer immunological memory for particular infectious pathogens.

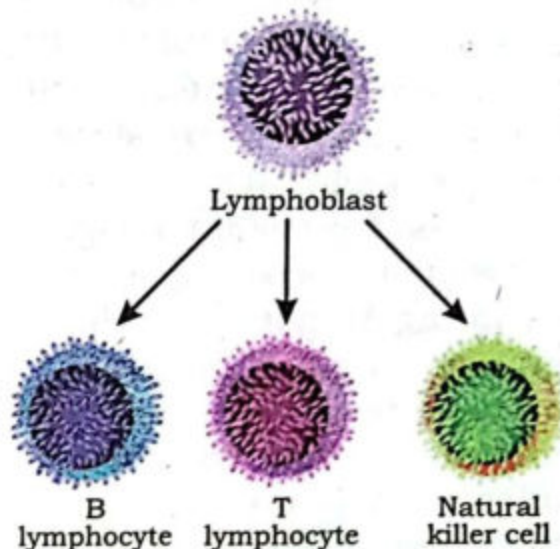


Fig: 13.12 Origin of different types of Lymphocytes from Lymphoblast (stem cell)

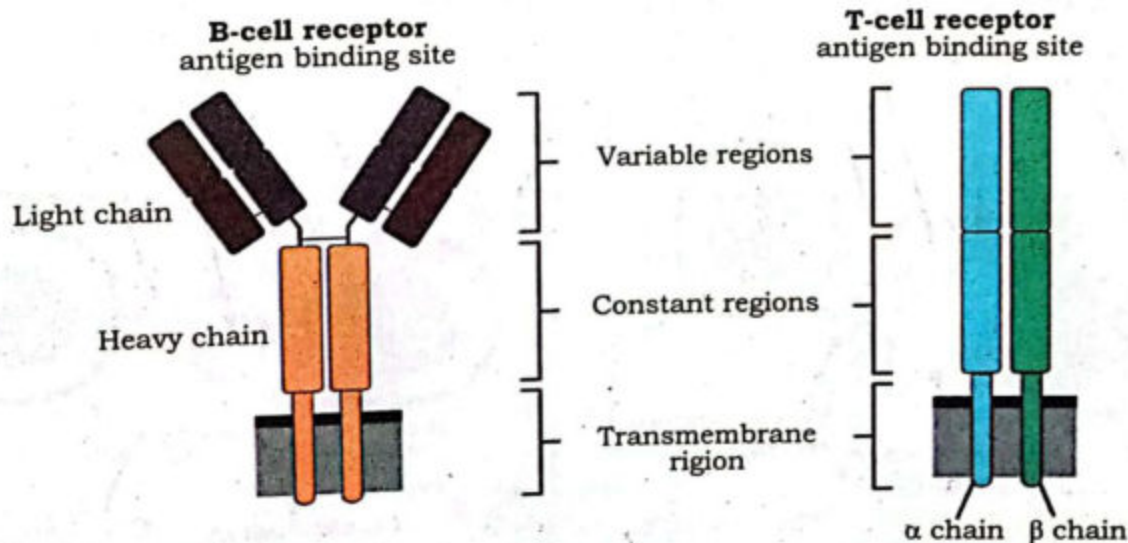


Fig: 13.13 Structure of B-Cells Receptor and T-Cells Receptor

T-Lymphocytes:

They are formed in bone marrow and then migrate to thymus to attain maturity. For this reason, they were named as T cells (T stands for thymus). A naïve T cells (inactive), in order to become an effector T Cell,

requires interaction with antigen presenting cell. They require essentially three signals; i.e., TCR, BCR and cytokines. According to the function and cell surface markers, T cells differentiate into four main types, i.e., T helper cells, T cytotoxic cells, T memory cells and T suppressor or regulatory cells.

T cytotoxic cell (T_c):

The T cells having CD8 markers (co-receptor) are termed as T_c . They are programmed to kill the target cells (apoptosis) that bears specific antigens. For this reason, they are termed as **Cytotoxic cells**. Their specific targets are virally infected cells, bacteria and tumor cells. The details of their activation and role are already discussed.

T helper cell (T_H):

As the name suggests, these cells upon activation by APCs, are involved in signaling by releasing various cytokines and thus ultimately activating B cells, T_c cells and other macrophages to destroy the specific antigen carrying pathogen. They play very important role our adaptive immune response. The T_H cells bear CD4 type (co-receptor) cell surface markers on their plasma membranes. Upon activation of T cells, it further differentiates into its sub-types which release different cytokines like Interleukins, INF, etc. for activation signals to other WBCs.

T suppressor or regulatory cell (T_s):

The T_s cells play vital role to suppress the activity of the activated T_c , T_H cells and B cells after they get rid of the specific antigen bearing pathogen, otherwise they may harm their own normal cells of the body. T_s cells have CD4, CD and TCR receptors on their surface.

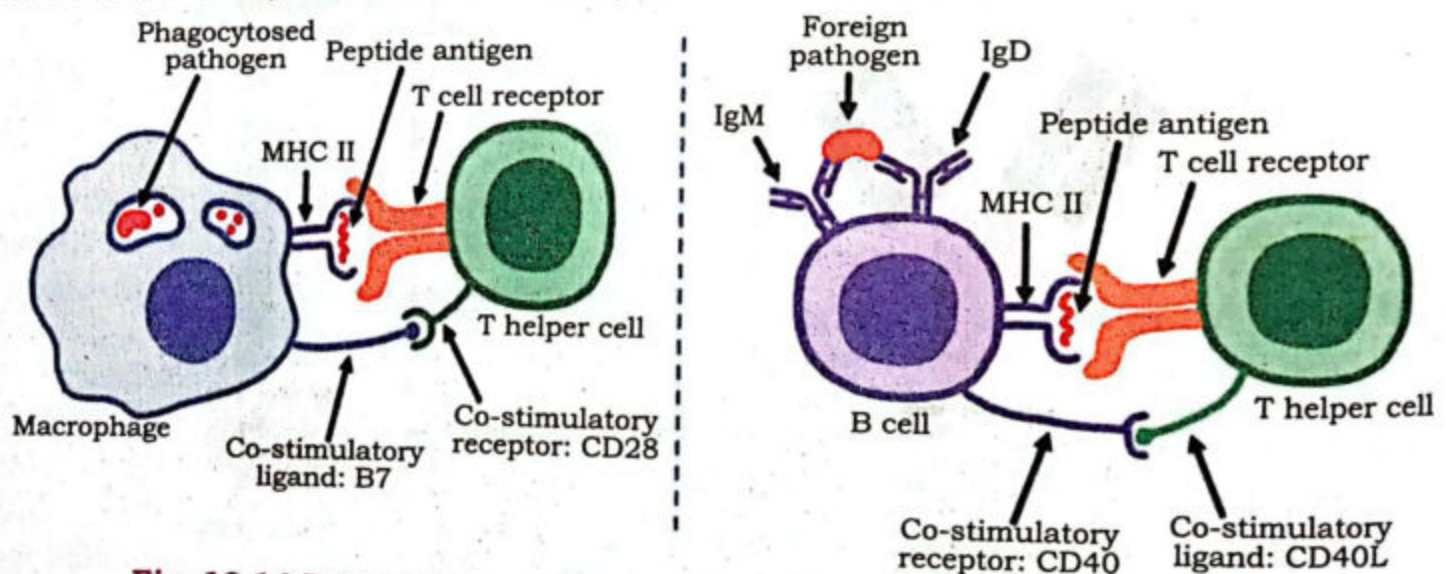


Fig: 13.14 Inter action and activation of T_H cell with B-Cell and antigen presenting cell (macrophage)

T memory cell (T_M):

A group of T cells that are antigen specific and capable to survive for years even after the elimination of infectious organism. As soon as they are re-exposed to the same antigen bearing organism, the T memory cells



rapidly proliferate and become effector T cells. In this way, they play extremely important role in controlling the reinfection. The process of vaccination is greatly benefitted by the presence of T_M cells.

B Lymphocytes:

These are formed and mature in the bone marrow. They confer upon us the most important aspect, the antibodies for our specific immunity. B Cells activation into effector cells called Plasma cell occurs either T-dependent mechanism or through T-Independent mechanisms. The former occurs when an activated T_H cell displays peptide antigen along with its MHC-II. Thus activated T_H cell releases the alert signals to naïve B-cell which then grows in size and starts producing and releasing antibodies actively. Such antibody secreting B-cells are termed as **Plasma cells**. Some of the B-cells meanwhile turn into **B Memory cells** or (B_M). The latter, records all the data of the antigen for controlling future reinfection. They do play very important role as far as the immunization is concerned.

① Cytotoxic T cell binds to infected cell

② Perforin makes holes in infected cell's membrane and enzyme enters

③ Infected cell is destroyed

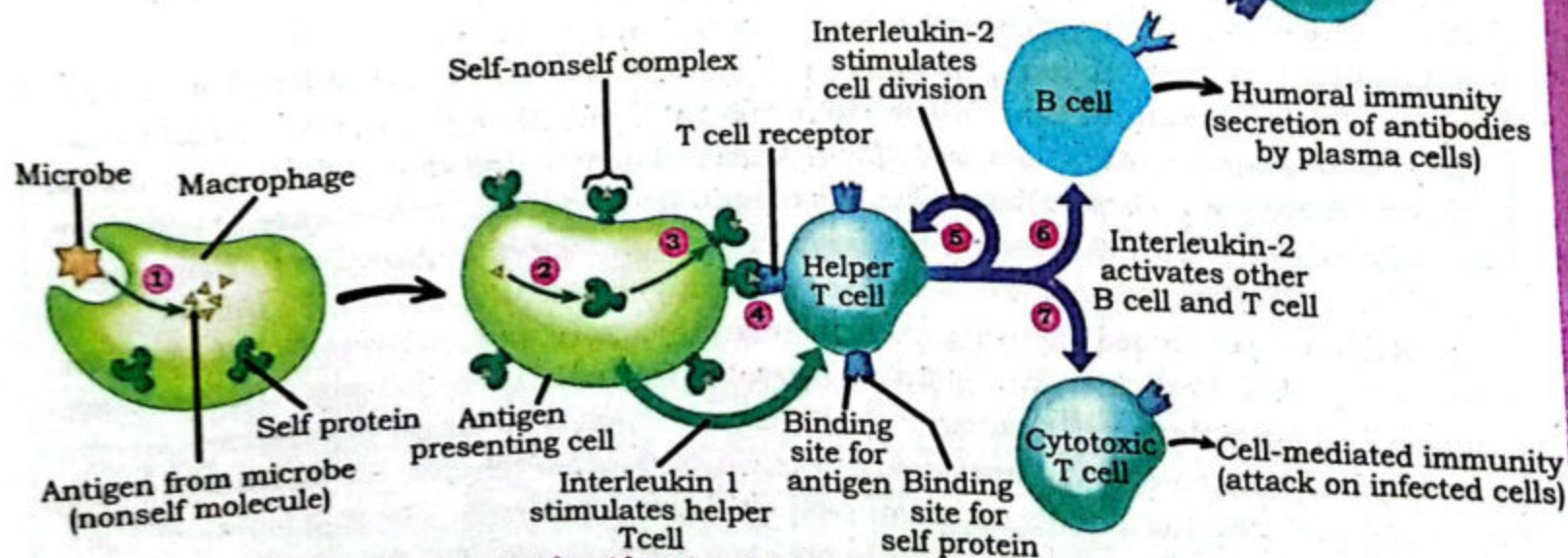
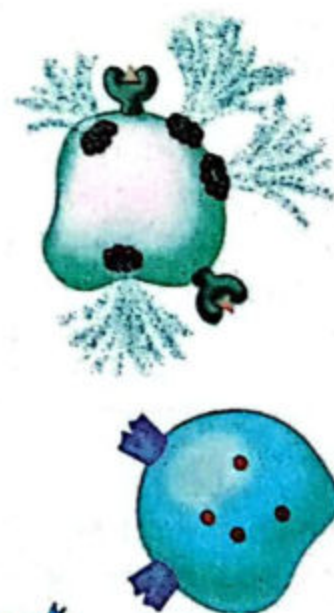
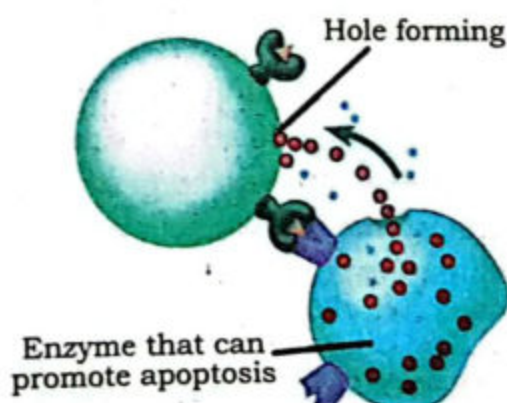
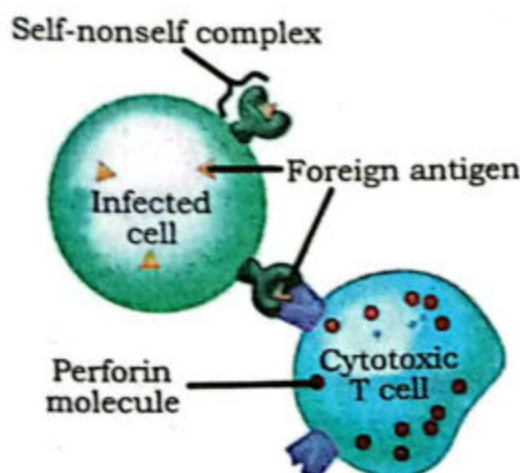
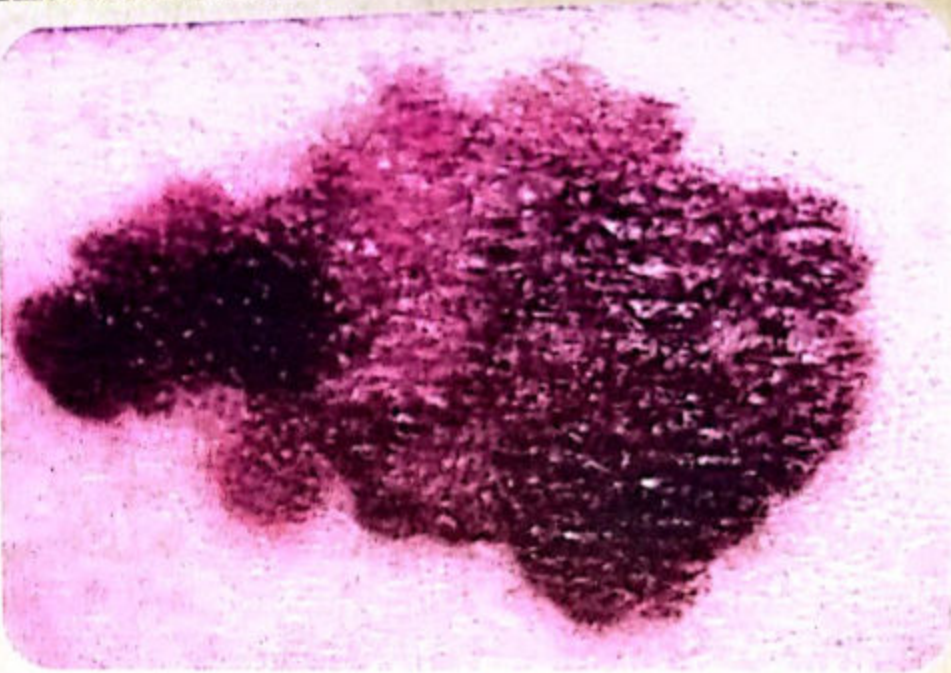


Fig: 13.15 Activation and class switching of B-cell

MALIGNANT MELANOMA

Malignant melanoma is a type of skin cancer of the melanin producing skin cells, the melanocytes. It is one of the most dangerous form of skin cancer. The usual cause for the development of melanoma is exposure to Ultra Violet radiation. However, genetic causes are also reported. In the early stage of melanoma development, the melanocytes show abnormally high, uncontrolled growth rate so spread radially in basal



epidermis. Then it starts moving upward in epidermis and this stage is the beginning of invasive stage when the cancerous cells may spread through body fluids to different parts of the body. At this stage, the Tumor Infiltrating Lymphocytes (TILs) recognize the abnormal behavior of the tissue and initiate immune response. TILs comprised of both T and B Lymphocytes along with NK cells, macrophages, dendritic cells, mast cells, basophils, eosinophils and neutrophils. The main proportion of TILs consists of Tc. In case of malignant melanoma TILs response is not strong enough to eliminate the cancerous cells. In such malignant melanoma micro-environment, number of molecules and cells have been shown to suppress immune response through cell interaction and/or the production of immune suppressing molecules.

MONOCLONAL ANTIBODIES

Antibodies derived from the clone of a single W.B.C. are termed as monoclonal antibodies or mAb. They bind only to the same epitope of the antigen. Monoclonal antibodies are being used to treat some types of cancer. They can be used alone or to carry drugs, toxins, or radioactive substances directly to cancer cells. Monoclonal antibodies were first produced in 1975 by Georges Köhler and César Milstein who were successful in making fusions of myeloma cell lines with B cells to create hybridomas that could produce antibodies, specific to known antigens and that were immortalized. They and Niels Kaj Jerne shared the Nobel Prize in Physiology or Medicine in 1984 for the discovery. George Winter and his team in 1928 developed techniques to make non-human antibodies compatible with human body. In 2018, James P. Allison and Tasuko Honjo succeeded in using mAb in treating cancer and awarded Nobel Prize for Physiology and Medicine. The simplest protocol for mAb is as follows: The mice are immunized with the desired antigen and after few days their blood is screened for the presence of antigen-specific antibodies. In case of positive results, the splenocytes are isolated and then fused with myeloma cells to make hybridoma. The myeloma cells are capable of unlimited replication. The hybridoma are screened for the specific-type and



class of antibodies they produce. The desired antibody producing clones are selected for further propagation. Thus the mAb are produced from cell culture techniques. In addition to the treatment of cancer, mAb are not only used for diagnosis of different diseases (e.g. hepatitis, etc) as they can identify particular antigens but also for therapeutic purposes as they can block the targeted substances inducing apoptosis of the target cells. They can also be used to prevent autoimmune diseases such as rheumatoid arthritis.

13.3.3 Disorders Of Immune System:

Allergies:

Sometimes, our immune system responds to certain normally harmless substances (e.g. pollens, molds, pet dander, certain drugs, some food, etc., called allergins. Such response is termed as allergy. It is considered to be an exaggerated response of the immune system against the normal substances taken by the WBCs as antigens. The common allergies of man are rhinitis, asthma, eczema, dermatitis, hives, anaphylaxis, food allergies, etc.

When allergins enter our body, a type of allergin specific antibody, IgE is produced by the plasma cells. These antibodies upon secretion into the blood bind to the allergin and then to the receptor on the mast cells or basophils. Being granulocytes, mast cells and basophils upon activation through IgE-allergin complex, release histamine to produce inflammatory response causing symptoms like itching, rashes, runny nose, sore throat, sneezing, cough, etc. The treatment of allergies usually employs anti-histamine and steroids.

Autoimmune Diseases:

It is the abnormality of the immune system in which immune system fails to recognize one or more own tissues as "self". As a result, it initiates immune response against its own cells. Diabetes mellitus type-I, rheumatoid arthritis, psoriasis, etc are some of the examples of autoimmune diseases in human. In such cases, misdirected self-antibodies are produced which destroy own tissue.

Autoimmune diseases can develop from a variety of factors with no known definitive cause for the onset of the disease. Some suspected sources are some bacterial or viral infections, effects from certain drugs, toxins and pollutants. In human, some of the common affected tissues are muscle, joints, blood vessels and connective tissues.

The common symptoms are malaise (sick feeling), low fever, fatigue, muscle ache and rashes on different areas of the body.

Justify why physicians prescribe antihistamine therapy to the patients of runny nose or skin rashes.

As you have already learnt that during the allergic reaction, mast cells and basophils release histamine to induce inflammation. In order to reduce inflammatory response,



antihistamine drugs are used to block the histamine receptors of mast cells, muscles, endothelial cells, etc. and ultimately the allergy is controlled. Itching, sneezing, runny nose, skin rashes are thus reduced in this way.

Transplant rejections:

Whenever some of the body organ either loses its function or lost due to some accident or incident, it is desired to replace it from corresponding healthy organ from a donor. Commonly transplanted organs are liver, kidney, skin and heart. However, transplanted organ is always subject to some degree of rejection because of fact that genetically donor and recipient are not identical. Thus the transplanted organ is subject to the attack of immune cells of the recipient, the response is termed as transplant rejection. It has been observed that different types of transplanted tissues encounter different rejection mechanisms of the host's immune system. It could be the consequence of CMI or humoral immunity.

Role of T-cells in transplant rejection:

In this case of transplant of solid organs from individual of the same species or allograft, the major threats are the T cells of the recipient. Initially, the dendritic cells of the donor tissue migrate into the recipient lymphoid tissue where they are recognized by the recipient's T_H cells through APCs. Once activated, T_H cells, proliferate and activate NK cells and T_C to attack the grafted tissue.

Role of B-cells in transplant rejection:

The activated T_H cells signal the B-cells to transform into specific antibody secreting Plasma Cells. The antibodies thus produced play very important role in destroying the grafted tissue.

Skills: Initiating and Planning

Explain why a transplant recipient is given immune suppressant drugs and determine what implications do this have on his life.

People having organ transplant are given immune suppressant drugs because of the threat of organ rejection by the immune cells of the recipient. These drugs suppress or inhibit the activity of the immune system. Different types of immune-suppressants are used for this purpose decided by the physician. There are no doubt, side effects of such drugs such as fatigue, headache, nausea, vomiting, acne, diabetes, hypertension, hair loss, mouth sore, sepsis, osteoporosis, etc.



SUMMARY

- Immune system is the defence system of an organs consists of cells and proteins.
- Study of functioning and disorders of immune system is termed as immunology.
- The non-specific type of immune system against any pathogen which tries to enter in the body called innate-immune system.
- Adaption immune system is the specific types of immune system develop against specific foreign body called antigen.
- Innate immune system is consist of 1st line of defence and 2nd line of defence
- 1st line of defence is consist of skin, mucous membrane, ciliated epithelium, saliva, tears, mucous, gastric juice etc.
- 2nd line of defence consist of phagocytes (neutrophils monocytes), dendritic cells, cytotoxic T cell, natural kill cells (N.K.C) perforin protein and granzyme.
- 3rd line of defence consist of lymphocyte T and B antibodies proteins.
- Active immunity is the consequence of individuals own immune response.
- Passive immunity is the kind of immunity. Where antibodies on immune cells produced by one individual are transferred to another individual to develop immunity.
- Allergies, auto immune diseases, transplant rejections are the disorder of immune system.

EXERCISE

1. **Encircle the correct choice**
- (i) The study of functioning and disorders of immune system is termed as:
(a) Immunity (b) Immune response
(c) Immunology (d) all of these
- (ii) It is responsible for specific immune response:
(a) Innate Immune system (b) First line of defense
(c) Second line of defense (d) Third line of defense
- (iii) Following serves as physical barrier:
(a) Macrophage (b) Mucous membrane
(c) HCl (d) Saliva
- (iv) Our digestive tract consists of what kind of barriers?
(a) Physical barriers (b) Biochemical barriers
(c) Both a & b (d) None of these



- (v) Which of the following is involved in phagocytosis of bacteria?
(a) Neutrophil (b) Macrophage
(c) Both a & b (d) None of these
- (vi) It is not involved in antigen presentation:
(a) Macrophage (b) Dendritic cell
(c) Neutrophil (d) Both a & b
- (vii) The type of receptor exhibited by all nucleated cells other than macrophages is:
(a) MHC-I (b) MHC-II (c) Both a & b (d) TCR
- (viii) Any foreign substance that can elicit immune response is called:
(a) Antibody (b) Antigen
(c) Both a & c (d) Antihistamine
- (ix) The type of immunity through the transfer of antibodies from mother to her fetus is:
(a) Natural Active Immunity (b) Natural Passive Immunity
(c) Artificial Active Immunity (d) Artificial Passive Immunity
- (x) Antibodies are secreted by:
(a) T cells (b) B cells
(c) Plasma cells (d) Macrophages

2. Write short answers of the following questions:

1. List out any six biochemical barriers.
2. How the tumor cells are dealt with by our immune system?
3. Differentiate between NK Cell and T_c cell.
4. List out the ways of second line of defense.
5. Even though the core proteins are the same, how antibodies differ from each other?
6. What is antipyretic therapy and why it is used?
7. List out four autoimmune disorders of man.
8. What is phagocytosis? Name some WBCs acting as phagocytes.
9. What is inflammation?
10. Outline the harmful effects of fever.

3. Write detailed answers of the following questions:

1. State and explain how a bacterium is identified as non-self by our immune system.
2. What is cytotoxicity? Compare and contrast the mechanisms of NK cell and T_c cell.
3. What is inflammation? Why it is developed and how it is controlled?
4. Explain the underlying mechanism of the fever. State and explain its protective role as well as harmful effects.
5. Discuss the structural features of antibody? How it is helpful in getting rid of pathogenic organisms.
6. What do you understand by autoimmune disorder? Explain the role of T cells and B cells in transplant rejection.